

First-Principles Wick-Contraction Derivations

Pion and Proton Measurements in pyquda_measurement_utils

Physics-audit technical note

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Abstract

This note derives the pion and proton measurement kernels in `pyquda_measurement_utils` from the target continuum matrix elements, through Euclidean spectral decompositions, flavor-labelled Grassmann integrals, propagator and sequential-source construction, and finally the spin-color contractions implemented by the production `einsum` expressions. It covers two-point functions, electromagnetic form factors (EMFF), quark and gluon energy-momentum tensor (EMT) building blocks, qTMD/qGTMD, qPDF/qGPD, pion qTMD wave functions and qDA, stochastic disconnected loops, the legacy pion soft-factor four-point function, and background-response measurements.

The central Wick results are that the pion backward line and trace are correct, the proton direct-exchange determinant and both flavor-specific sequential kernels are correct, and the qTMDWF/qDA endpoint orientation is correct. A physical sign is missing when the saved positive stochastic trace in the disconnected qTMD/PDF workflow is interpreted as $\bar{q}\Gamma Wq$. Most remaining limitations separate a raw Euclidean correlator from a renormalized and matched physical observable; they are not additional Wick-contraction failures.

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1 Scope, baseline, and meaning of “measurement”

1.1 Audited source and document purpose

All formula-to-code statements in this document refer to repository commit 52927d03f30c (the full hash appears on the title page). The principal production files are:

- `pion_utils_vibe_develop.py`, `proton_utils_vibe_develop.py`, and `bw_seq_pyquda.py`;
- `pion_EMFF_vibe_develop.py`, `pion_EMT_vibe_develop.py`, and `proton_EMT_vibe_develop.py`;
- `pion_qTMD_vibe_develop.py`, `proton_qTMD_pyquda.py`, `pion_qTMDWF_pyquda.py`, and `qTMD_operator_utils.py`;
- `Disconnected_1pt_qTMD_vibe_develop.py`, `Disconnected_1pt_EMT_vibe_develop.py`, and `flowed_quark_ringed_norm.py`;
- `pion_soft_factor_vibe_develop.py` and `pion_current_background_response_vibe_develop.py`.

This is a derivation and audit note, not an analysis prescription with a single universal ratio. The source and sink interpolators, polarization projectors, temporal boundary effects, excited states, and the renormalization scheme must be retained in an ensemble analysis.

1.2 Three distinct output levels

The word observable is often used for three different objects:

1. a *raw Euclidean correlator*, such as C_2 , C_3 , or a stochastic one-point trace;
2. a *bare lattice matrix element*, obtained after a spectral fit or ratio removes overlaps and Euclidean exponentials;
3. a *physical observable*, after charge/flavor combination, improvement, renormalization, soft subtraction, continuum matching, scale evolution, and, for the flowed EMT, the small-flow-time expansion.

The audited measurement code predominantly produces level 1. In correctly chosen kinematics its output can be used to obtain level 2. None of EMFF, EMT, qTMD/qGTMD, qPDF/qGPD, qTMDWF, or qDA is automatically a level-3 observable in the current producer.

Measurement	Continuum target	Object directly produced
Pion EMFF	$\langle \pi(p_f) J_\mu^{\text{em}} \pi(p_i) \rangle$ and $F_\pi(Q^2)$	One-flavor connected local-current C_3 , without the full charge/flavor combination, Z_V , improvement, disconnected term, or ratio.
Proton EMFF	$F_1(Q^2)$ and $F_2(Q^2)$	No dedicated producer/extraction; the local limit of the common proton insertion kernel is only the raw connected building block.
Quark EMT	$\langle H(p_f) T_{\mu\nu}^q H(p_i) \rangle$ and gravitational form factors	Connected and disconnected flowed local/one-derivative primitives.

Measurement	Continuum target	Object directly produced
Gluon EMT	$\langle H(p_f) T_{\mu\nu}^g H(p_i) \rangle$	Flowed $2 \text{Tr}_c \sum_\rho F_{\mu\rho} F_{\nu\rho}$, not a complete matched EMT.
qTMD/ qGTMD	Staple-bilocal hadron matrix element and its factorized light-cone quantity	Bare coordinate-space connected or stochastic trace correlator.
qPDF/qGPD	Straight-bilocal quasi distribution/generalized distribution	Bare coordinate-space connected or stochastic trace correlator.
qTMDWF/ qDA	Vacuum-to-pion bilocal wave-function/distribution-amplitude matrix element	Bare vacuum-to-pion correlator; qTMDWF is currently CG/fixed-gauge.
Soft factor	Soft subtraction compatible with the chosen TMD factorization scheme	A legacy pion wall-source four-propagator C_4 , whose universality for the GI staple scheme is not established by the contraction alone.
Response	Derivatives of correlators with respect to external operator couplings	Ordered SOS and SO_2SO_1S response propagators for diagnostics.

2 Euclidean, array, Fourier, and spectral conventions

2.1 Gamma matrices and propagator orientation

The code uses the PyQUDA/DeGrand–Rossi Euclidean basis,

$$\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}, \quad \gamma_\mu^\dagger = \gamma_\mu, \quad \gamma_5 = \gamma_1\gamma_2\gamma_3\gamma_4, \quad (2.1)$$

and the Wilson–clover operator satisfies

$$D^\dagger = \gamma_5 D \gamma_5, \quad S(y, x) = D^{-1}(y, x) = \gamma_5 S(x, y)^\dagger \gamma_5. \quad (2.2)$$

For one flavor,

$$S_{\alpha\beta}^{ab}(x, y) = \langle q_\alpha^a(x) \bar{q}_\beta^b(y) \rangle. \quad (2.3)$$

The production propagator layout is

```
prop.data[w,t,z,y,x_cb, spin_sink, spin_source,
          color_sink, color_source]
```

so the final four indices are ordered as $S_{\alpha\beta}^{ab}$. A Hermitian conjugate must exchange *both* spin and color endpoints.

The common pion backward-line expression is

```
einsum("ij,wtzyxlab,kl->wtzyxkjb",
       gamma5, prop.data.conj(), gamma5)
```

and therefore returns

$$\begin{aligned} B_{\delta\gamma}^{ba}(x) &= (\gamma_5)_{\delta\ell} [S_{\rho\ell}^{ab}(x, x_0)]^* (\gamma_5)_{\rho\gamma} \\ &= S_{\delta\gamma}^{ba}(x_0, x). \end{aligned} \quad (2.4)$$

This is a full compound spin–color dagger, not a color-only transpose.

2.2 Propagator construction and smearing

For flavor f ,

$$D_f S_f(\cdot, x_0) = \eta_{x_0}, \quad S_f = D_f^{-1}. \quad (2.5)$$

Writing a spatial smearing operator as $\mathcal{S}_k$, the source- and sink-smeared line has the schematic form

$$S_f^{XY}(x, x_0) = \mathcal{S}_X(x) D_f^{-1}(x, x_0) \mathcal{S}_Y^\dagger(x_0). \quad (2.6)$$

Momentum-boosted smearing modifies overlap with a moving state. It does not replace the Fourier momentum projection and by itself does not set the physical hadron momentum. The positive- and negative-boost pion lines may be chosen as active or spectator lines, but flavor labels must be assigned before the isospin-degenerate equality $S_u = S_d$ is used.

2.3 Fourier phases and the actual external kinematics

`MomentumPhase` constructs

$$\Phi_n(x; x_0) = \exp \left[+2\pi i \sum_{j=1}^3 \frac{n_j(x_j - x_{0j})}{L_j} \right]. \quad (2.7)$$

Two-point applications pass $-\mathbf{p}$, which gives the standard $\exp[-i\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}_0)]$ projector. The meson or baryon fixed-sink sequential source contains $\Phi_{\mathbf{p}_f}(y)$. The subsequent complex conjugation used to form the backward sequential line turns it into the effective sink projector $\exp[-i\mathbf{p}_f \cdot (\mathbf{y} - \mathbf{x}_0)]$. The insertion is multiplied by $\Phi_{\mathbf{q}}(x)$.

Before projection, translation covariance gives

$$C_3(y, x; x_0) \sim e^{+i\mathbf{p}_f \cdot (\mathbf{y} - \mathbf{x})} e^{+i\mathbf{p}_i \cdot (\mathbf{x} - \mathbf{x}_0)}. \quad (2.8)$$

Multiplication by the two code phases leaves a spatial sum only if

$$\boxed{\mathbf{q} = \mathbf{p}_f - \mathbf{p}_i, \quad \mathbf{p}_i = \mathbf{p}_f - \mathbf{q}.} \quad (2.9)$$

The module docstrings and dedicated measurement notes use the executed plus-sign insertion convention in Eq. (2.7). A sequential source is built with $+\mathbf{p}_f$; complex conjugation in the backward line produces the effective sink projector with $-\mathbf{p}_f$. The numerical kinematics is Eq. (2.9).

For off-forward bilocals,

$$\Delta = p_f - p_i = q, \quad P = \frac{p_f + p_i}{2} = p_f - \frac{q}{2}, \quad (2.10)$$

$$t = \Delta^2 = (E_f - E_i)^2 - |\Delta|^2, \quad \xi = -\frac{\Delta^+}{2P^+}. \quad (2.11)$$

Thus a nonzero $\mathbf{q}$ is qGTMD/qGPD kinematics, not a forward qTMD/qPDF measurement.

2.4 Spectral decompositions

With relativistically normalized finite-volume states,

$$C_2^H(t, \mathbf{p}) = \sum_n \frac{Z_n(\mathbf{p}) Z_n^\dagger(\mathbf{p})}{2E_n} e^{-E_n t} \text{backward/wrap contributions}. \quad (2.12)$$

For $0 < \tau < t_{\text{sep}}$,

$$C_3^H(\tau, t_{\text{sep}}; p_f, p_i) \longrightarrow \frac{Z_f(\mathbf{p}_f) Z_i^\dagger(\mathbf{p}_i)}{4E_f E_i} e^{-E_f(t_{\text{sep}} - \tau)} e^{-E_i \tau}$$

$$\times \langle H(p_f) | \mathcal{O} | H(p_i) \rangle. \quad (2.13)$$

The producer does not perform the spectral fit or construct a complete form-factor/distribution ratio. An antiperiodic quark temporal boundary does not remove hadron wrap-around terms.

For a vacuum-to-pion bilocal,

$$\begin{aligned} C_\Gamma(t, \mathbf{P}, b) &= \sum_{\mathbf{x}} e^{-i\mathbf{P} \cdot (\mathbf{x} - \mathbf{x}_0)} \langle \mathcal{O}_\Gamma(\mathbf{x}, t; b) \mathcal{O}_\pi^\dagger(\mathbf{x}_0) \rangle \\ &\longrightarrow \frac{\langle 0 | \mathcal{O}_\Gamma(b) | \pi(P) \rangle Z_\pi^\dagger(\mathbf{P})}{2E_\pi} e^{-E_\pi t}. \end{aligned} \quad (2.14)$$

3 Grassmann integration and universal loop signs

3.1 Generating functional

For one flavor and sources $\eta, \bar{\eta}$,

$$\begin{aligned} Z[\eta, \bar{\eta}] &= \int Dq D\bar{q} \exp[-\bar{q} D q + \bar{\eta} q + \bar{q} \eta] \\ &= \det D \exp[\bar{\eta} D^{-1} \eta]. \end{aligned} \quad (3.1)$$

Differentiating in a fixed source order gives

$$\langle q_i \bar{q}_j \rangle = S_{ij}, \quad \langle q_i q_k \bar{q}_j \bar{q}_l \rangle = S_{ij} S_{kl} - S_{il} S_{kj}. \quad (3.2)$$

Equation (3.2) is a determinant over complete endpoint labels. If $i = (\alpha, a)$ is a spin-color compound index, exchange means exchanging the full pair (α, a) .

3.2 Closed-fermion-loop minus sign

Introduce a bilinear perturbation:

$$Z[\lambda] = \int Dq D\bar{q} e^{-\bar{q}(D + \lambda M)q} = \det(D + \lambda M). \quad (3.3)$$

Then

$$\left. \frac{\partial \ln Z}{\partial \lambda} \right|_0 = \text{Tr}(D^{-1} M) = -\langle \bar{q} M q \rangle, \quad (3.4)$$

and hence

$$\boxed{\langle \bar{q} M q \rangle = -\text{Tr}[MS]}. \quad (3.5)$$

For noise satisfying $\mathbb{E}[\xi_i \xi_j^*] = \delta_{ij}$, let $\eta = D^{-1} \xi$. The stochastic identity is

$$\mathbb{E}_\xi[\xi^\dagger M \eta] = \text{Tr}[M D^{-1}]. \quad (3.6)$$

Equation (3.6) estimates a positive matrix trace; it does not generate the physical closed-loop minus in Eq. (3.5).

4 Pion two- and three-point Wick contractions

4.1 Pion interpolators and complete C2 derivation

Use a charged-pion sink

$$\mathcal{O}_s(x) = \bar{d}_\gamma^c(x) (\Gamma_s)_{\gamma\alpha} u_\alpha^c(x) \quad (4.1)$$

and source

$$\mathcal{O}_0^\dagger(x_0) = \bar{u}_\beta^b(x_0)(\Gamma_0)_{\beta\delta}d_\delta^b(x_0). \quad (4.2)$$

The field order is

$$\bar{d}(x), u(x), \bar{u}(x_0), d(x_0). \quad (4.3)$$

Flavor conservation permits only

$$u_\alpha^c(x)\bar{u}_\beta^b(x_0) \longrightarrow (S_u)_{\alpha\beta}^{cb}(x, x_0), \quad (4.4)$$

$$d_\delta^b(x_0)\bar{d}_\gamma^c(x) \longrightarrow (S_d)_{\delta\gamma}^{bc}(x_0, x). \quad (4.5)$$

Even if $m_u = m_d$, pairings such as $u\bar{d}$ vanish because the Dirac operator is flavor diagonal. Moving the Grassmann variables into paired order gives the connected-meson sign,

$$C_{2,\text{formal}}^\pi(x) = -\text{Tr}_{s,c}[S_d(x_0, x)\Gamma_s S_u(x, x_0)\Gamma_0]. \quad (4.6)$$

For the Euclidean pseudoscalar,

$$(\bar{d}\gamma_5 u)^\dagger = \bar{u}\gamma_4\gamma_5^\dagger\gamma_4 d = -\bar{u}\gamma_5 d. \quad (4.7)$$

This adjoint minus cancels the Grassmann minus. With the production source convention $\Gamma_{\text{src}} = +\gamma_5$, the implemented correlator is

$$C_2^\pi(x) = \text{Tr}_{s,c}[S_d(x_0, x)\gamma_5 S_u(x, x_0)\gamma_5]. \quad (4.8)$$

Using Eq. (2.2), equal degenerate lines give

$$C_2^\pi(x) = \text{Tr}_{s,c}[S(x, x_0)^\dagger S(x, x_0)] \geq 0. \quad (4.9)$$

4.2 Index map to the pion production contraction

Let the backward line returned by Eq. (2.4) be B_{ji}^{cf} . The production operations are

$$X_{jm}^{cf} = B_{ji}^{cf}(\Gamma_s)_{im}, \quad (4.10)$$

$$C_{\text{site}} = X_{ji}^{ab}(S_u)_{il}^{ba}(\Gamma_{\text{src}})_{lj}. \quad (4.11)$$

The actual expressions are:

```
sink_inserted =
    einsum("wtzyxjicf,im->wtzyxjmcf",
           backward_line, sink_gamma)
corr_site =
    einsum("wtzyxjiab,wtzyxilba,lj->wtzyx",
           sink_inserted, forward_prop, source_gamma)
```

Every spin and color index appears exactly twice, in the Wick order of Eq. (4.8). The final contraction with "qwtzyx,wtzyx->qt" applies the Fourier projector and sums the spatial sites.

4.3 Connected pion C3 with a bilocal insertion

For an active- u operator based at x ,

$$J_\Gamma^u(x; b) = \bar{u}(x)\Gamma\mathcal{W}[x, x+b]u(x+b), \quad (4.12)$$

the flavor-preserving connected Wick path is

$$C_{3,u}^\pi = \sum_{x,y} e^{-ip_f \cdot (y-x_0)} e^{iq \cdot (x-x_0)} \times \text{Tr}_{s,c}[S_d(x_0, y)\Gamma_{\text{sink}}S_u(y, x)\Gamma\mathcal{W}[x, x+b]S_u(x+b, x_0)\Gamma_{\text{src}}]. \quad (4.13)$$

The antiquark d line is the spectator. An insertion on the d line has a separately flavor-labelled path; isospin symmetry may relate the two only after both paths have been defined.

4.4 Meson fixed-sink sequential source

The sink sum in Eq. (4.13) is absorbed into a sequential solve. With the spectator source-to-sink propagator, the source placed on $y_4 = t_0 + t_{\text{sep}}$ is

$$\eta_{\text{seq}}(y) = \delta_{y_4, t_0 + t_{\text{sep}}} \Phi_{\mathbf{p}_f}(y; x_0) \left[\gamma_5 \Gamma_{\text{sink}}^\dagger \gamma_5 \right] S_{\text{spectator}}(y, x_0). \quad (4.14)$$

The code constructs

$$\Gamma_{\text{seq}} = \gamma_5 \Gamma_{\text{sink}}^\dagger \gamma_5 \quad (4.15)$$

with

```
gamma_seq = einsum("ab,cb,cd->ad",
                  gamma5, sink_gamma.conj(), gamma5)
```

and solves

$$DS_{\text{seq}} = \eta_{\text{seq}}. \quad (4.16)$$

The backward sequential line

$$B_{\text{seq}}(x) = \gamma_5 S_{\text{seq}}(x)^\dagger \gamma_5 \quad (4.17)$$

contains the conjugated sink phase and the summed line $S_d(x_0, y) \Gamma_{\text{sink}} S_u(y, x)$. The remaining local trace is

$$\text{Tr}_{s,c} [B_{\text{seq}}(x) \Gamma \mathcal{W}[x, x+b] S_{\text{active}}(x+b, x_0) \Gamma_{\text{src}}], \quad (4.18)$$

which is precisely Eq. (4.13).

5 Proton Wick determinant and sequential cuts

5.1 Interpolating field

Write the proton sink as

$$\chi_m(x) = \epsilon_{abc} G_{ij} u_i^a(x) d_j^b(x) u_m^c(x), \quad G = C \Gamma_{\text{interp}}, \quad (5.1)$$

with $C = i\gamma_2\gamma_4$. Production supports $C\gamma_5$, $C\gamma_4\gamma_5$, and $C\gamma_3\gamma_5$, labelled **5**, **T5**, and **Z5**. All supported G matrices are antisymmetric in the adopted basis,

$$G^T = -G. \quad (5.2)$$

The projected source contains

$$\bar{\chi}_n(x_0) \propto \epsilon_{def} G_{kl} \bar{u}_k^d(x_0) \bar{d}_l^e(x_0) \bar{u}_n^f(x_0), \quad (5.3)$$

where the fixed Euclidean-conjugation convention is absorbed in G and the external spin projector P_{mn} . For positive-parity unpolarized projection, $P_+ = (1 + \gamma_4)/4$.

5.2 Flavor-labelled Wick expansion

The d flavor has the unique pairing

$$d_j^b(x) \bar{d}_l^e(x_0) \longrightarrow (S_d)_{jl}^{be}. \quad (5.4)$$

The two identical u fields give the determinant

$$\det \begin{pmatrix} (S_u)_{ik}^{ad} & (S_u)_{in}^{af} \\ (S_u)_{mk}^{cd} & (S_u)_{mn}^{cf} \end{pmatrix} = (S_u)_{ik}^{ad} (S_u)_{mn}^{cf} - (S_u)_{in}^{af} (S_u)_{mk}^{cd}. \quad (5.5)$$

The exchange in the second product swaps complete spin–color source endpoints. Including the common field-order convention used by production,

$$\boxed{\mathcal{C}_p[S_u, S_d] = -\mathcal{D} + \mathcal{X}}, \quad (5.6)$$

where

$$\mathcal{D} = \epsilon_{abc}\epsilon_{def}G_{ij}G_{kl}P_{mn}(S_u)_{ik}^{ad}(S_d)_{jl}^{be}(S_u)_{mn}^{cf}, \quad (5.7)$$

$$\mathcal{X} = \epsilon_{abc}\epsilon_{def}G_{ij}G_{kl}P_{mn}(S_u)_{in}^{af}(S_d)_{jl}^{be}(S_u)_{mk}^{cd}. \quad (5.8)$$

The equality $S_u = S_d$ may be imposed for degenerate masses after Eqs. (5.7)–(5.8) are derived. It does not create an u -to- $\bar{d}$ contraction.

5.3 Why the printed second term is the complete exchange

The scalar represented by the second production `einsum` is

$$T_2^{\text{code}} = \epsilon_{abc}\epsilon_{def}G_{ij}G_{kl}P_{mn}S_{ik}^{ad}S_{jn}^{be}S_{ml}^{cf}. \quad (5.9)$$

The middle propagator in Eq. (5.9) can look like a partial u/d exchange if the dummy labels are interpreted as persistent flavor labels. They are not. For a single degenerate propagator tensor, simultaneous relabelling of source spin and color dummy indices through both epsilon tensors, followed by $G^T = -G$, yields

$$T_2^{\text{code}} = -\mathcal{X} \quad (S_u = S_d = S). \quad (5.10)$$

Consequently the production combination

```
corr = -term1 - term2
```

is

$$-\mathcal{D} - (-\mathcal{X}) = -\mathcal{D} + \mathcal{X}. \quad (5.11)$$

An independent explicit compound-index sum found

$$|T_2^{\text{code}} + \mathcal{X}| = 5.66 \times 10^{-14}, \quad (5.12)$$

$$|C_{\text{code}} - (-\mathcal{D} + \mathcal{X})| = 5.55 \times 10^{-14} \quad (5.13)$$

for random complex propagators. A spin-only or color-only swap does not satisfy this identity.

5.4 Flavor-specific sequential kernels as derivatives

The mathematically unambiguous definition of a cut baryon line is the holomorphic derivative of the same flavor-labelled polynomial:

$$(K_d)_{\alpha\beta}^{ab} = \frac{\partial \mathcal{C}_p[S_u, S_d]}{\partial (S_d)_{\alpha\beta}^{ab}}, \quad (K_u)_{\alpha\beta}^{ab} = \frac{\partial \mathcal{C}_p[S_u, S_d]}{\partial (S_u)_{\alpha\beta}^{ab}}. \quad (5.14)$$

Because $\mathcal{C}_p$ is linear in S_d , K_d has two terms: the derivative of the direct and exchange products. Because it is quadratic in S_u , K_u has four terms: both u factors can be cut in each of the direct and exchange products. This explains the two terms in `down_quark_insertion_pyquda` and the four terms in `up_quark_insertion_pyquda`; the latter already includes the multiplicity of the two identical u lines.

The line converted into a Dirac source has reversed compound endpoints:

$$K_{\text{source}}[\beta, \alpha, b, a] = \frac{\partial \mathcal{C}_p}{\partial S[\alpha, \beta, a, b]}. \quad (5.15)$$

Both the spin and color pairs are transposed. Exact polynomial differentiation gives maximum discrepancies

$$\|K_d^{\text{code}} - K_d^{\text{exact}}\|_{\infty} = 1.99 \times 10^{-15}, \quad \|K_u^{\text{code}} - K_u^{\text{exact}}\|_{\infty} = 2.08 \times 10^{-14}. \quad (5.16)$$

The production-adaptor comparison gives approximately 3.6×10^{-15} for each flavor.

5.5 Proton C3 after the sequential solve

After inverting the flavor-specific sequential source, a local or displaced insertion has the common form

$$C_{3,f} = \sum_x e^{+iq \cdot (x-x_0)} \text{Tr}_{s,c} [B_{\text{seq},f}(x) \Gamma \mathcal{W}[x, x+b] S_f(x+b, x_0)]. \quad (5.17)$$

The connected nucleon runner uses

```
einsum("pwtzyxjicf,gim,wtzyxmjfc->pgwtzyx",
       sequential, gamma_ls, current_prop)
```

which maps exactly to

$$(B_{\text{seq},f})_{ji}^{cf} \Gamma_{im} (\mathcal{W} S_f)_{mj}^{fc}. \quad (5.18)$$

The same cut flavor closes both spin and color. No additional factor two may be applied to the up channel in downstream analysis.

6 Electromagnetic form factors

6.1 Continuum matrix elements

The electromagnetic current is

$$J_{\mu}^{\text{em}} = \sum_f e_f \bar{q}_f \gamma_{\mu} q_f. \quad (6.1)$$

For a charged pion,

$$\langle \pi(p_f) | J_{\mu}^{\text{em}} | \pi(p_i) \rangle = (p_f + p_i)_{\mu} F_{\pi}(Q^2), \quad Q^2 = -q^2. \quad (6.2)$$

For the proton,

$$\langle p(p_f, s_f) | J_{\mu}^{\text{em}} | p(p_i, s_i) \rangle = \bar{u}(p_f, s_f) \left[\gamma_{\mu} F_1(Q^2) + \frac{i \sigma_{\mu\nu} q_{\nu}}{2M_N} F_2(Q^2) \right] u(p_i, s_i). \quad (6.3)$$

Solving Eq. (6.3) for F_1, F_2 requires several current directions, external momenta, and spin projectors. The Sachs form factors are

$$G_E = F_1 - \frac{Q^2}{4M_N^2} F_2, \quad G_M = F_1 + F_2. \quad (6.4)$$

For Wilson–clover fermions, a local current generally enters as

$$J_{\mu,R}^f = Z_V^f (1 + b_V a m_f) [\bar{q}_f \gamma_{\mu} q_f + a c_V \partial_{\nu} (\bar{q}_f \sigma_{\mu\nu} q_f)], \quad (6.5)$$

with scheme/action-dependent coefficients. Neither the flavor charges in Eq. (6.1) nor the factors in Eq. (6.5) can be inferred from a gamma insertion.

6.2 Pion local-current contraction

Setting $b = 0$, $\mathcal{W} = 1$, and $\Gamma = \gamma_\mu$ in Eq. (4.13) gives

$$C_{3,u}^{\pi,\mu} = \sum_{x,y} e^{-i\mathbf{p}_f \cdot (\mathbf{y} - \mathbf{x}_0)} e^{+i\mathbf{q} \cdot (\mathbf{x} - \mathbf{x}_0)} \times \text{Tr}_{s,c} [S_d(x_0, y) \Gamma_{\text{sink}} S_u(y, x) \gamma_\mu S_u(x, x_0) \Gamma_{\text{src}}]. \quad (6.6)$$

`pion_EMFF_vibe_develop.py` computes Eq. (6.6) through the common 16-gamma scan and the meson sequential line. The phase relation is $\mathbf{p}_i = \mathbf{p}_f - \mathbf{q}$.

This quantity is a one-active-line raw connected C_3 , not $F_\pi(Q^2)$. A complete analysis must:

1. include $C_2(\mathbf{p}_f)$ and $C_2(\mathbf{p}_i)$ with matching interpolators/smearing;
2. cancel overlap factors and exponentials or perform a simultaneous spectral fit;
3. combine the active- u and active- d contractions with e_u and e_d , and add any required disconnected contribution;
4. apply current improvement and Z_V ;
5. divide by the kinematic factor in Eq. (6.2).

The production application sets its two-point momentum list from `qext`. For nonzero $\mathbf{p}_f$, that list need not contain both $\mathbf{p}_f$ and $\mathbf{p}_f - \mathbf{q}$. Existing C_3 can remain usable, but the missing matching C_2 must then be calculated separately.

6.3 Proton local-current status

The proton local-current connected building block is obtained from Eq. (5.17) at $b = 0$ with $\Gamma = \gamma_\mu$, separately for $f = u, d$:

$$C_{3,\mu}^{p,\text{conn}} = e_u C_{3,u}^\mu + e_d C_{3,d}^\mu. \quad (6.7)$$

The common direct-exchange and sequential kernels are suitable for this construction. At the audited commit there is, however, no dedicated proton EMFF producer that defines the complete local-current charge combination, projector set, kinematic linear system, and F_1/F_2 extraction. A proton qTMD file at $b = 0$ must not automatically be labelled F_1 or F_2 .

7 Quark and gluon energy-momentum tensor measurements

7.1 Continuum operators and hadron decompositions

The symmetric quark one-derivative operator used as the lattice target is

$$\mathcal{O}_{\mu\nu}^q = \frac{1}{4} \bar{q} \left[\gamma_\mu \overleftrightarrow{D}_\nu + \gamma_\nu \overleftrightarrow{D}_\mu \right] q, \quad \overleftrightarrow{D}_\nu = \overrightarrow{D}_\nu - \overleftarrow{D}_\nu. \quad (7.1)$$

Flavor singlet quark operators mix with gluon operators under renormalization.

For a spin-zero pion, one common decomposition is

$$\begin{aligned} \langle \pi(p_f) | T_{\mu\nu} | \pi(p_i) \rangle &= 2P_\mu P_\nu A_\pi(t) + \frac{1}{2} \left(\Delta_\mu \Delta_\nu - \delta_{\mu\nu} \Delta^2 \right) D_\pi(t) \\ &\quad + 2m_\pi^2 \delta_{\mu\nu} \bar{C}_\pi(t), \end{aligned} \quad (7.2)$$

subject to Euclidean continuation and operator-basis conventions. For a spin- $\frac{1}{2}$ proton, a standard form is

$$\begin{aligned} \langle p_f | T_{\mu\nu} | p_i \rangle = & \bar{u}(p_f) \left[A(t) \gamma_{\{\mu} P_{\nu\}} + B(t) \frac{i P_{\{\mu} \sigma_{\nu\}} \Delta_\rho}{2M_N} \right. \\ & \left. + D(t) \frac{\Delta_\mu \Delta_\nu - \delta_{\mu\nu} \Delta^2}{4M_N} + \bar{C}(t) M_N \delta_{\mu\nu} \right] u(p_i). \end{aligned} \quad (7.3)$$

Braces denote symmetrization with the convention chosen in the analysis. The measurement code does not solve either gravitational-form-factor system.

7.2 Central gauge-covariant derivative

The lattice shift convention is

$$(\mathcal{T}_{+\mu} q)(x) = U_\mu(x) q(x + \hat{\mu}), \quad (\mathcal{T}_{-\mu} q)(x) = U_\mu^\dagger(x - \hat{\mu}) q(x - \hat{\mu}). \quad (7.4)$$

The dimensionless central derivative implemented by `_covdev_sym_prop` is

$$aD_\mu q(x) = \frac{1}{2} \left[U_\mu(x) q(x + \hat{\mu}) - U_\mu^\dagger(x - \hat{\mu}) q(x - \hat{\mu}) \right]. \quad (7.5)$$

Let B denote the sequential/backward line and S the active line. For a nonsymmetrized gamma-derivative pair, production evaluates

$$\mathcal{B}_{\mu\nu} = \frac{1}{2} B \gamma_\mu (D_\nu S) - \frac{1}{2} (B \overleftarrow{D}_\nu) \gamma_\mu S. \quad (7.6)$$

It then forms

$$\mathcal{O}_{\mu\nu}^{\text{code}} = \frac{1}{2} (\mathcal{B}_{\mu\nu} + \mathcal{B}_{\nu\mu}). \quad (7.7)$$

The two halves in Eq. (7.6) and the final half in Eq. (7.7) reproduce the 1/4 in Eq. (7.1), in lattice units. No $1/a$ is inserted. There is no automatic spacetime trace subtraction.

7.3 Connected pion EMT contraction

At flow time t_f , denote the flowed active and sequential lines by $S(t_f)$ and $B(t_f)$. The primitive local field is

$$L_\Gamma(x; t_f) = \text{Tr}_{s,c} [B(x; t_f) \Gamma S(x; t_f) \Gamma_{\text{src}}]. \quad (7.8)$$

The derivative primitive is

$$L_{\Gamma,\nu}(x; t_f) = \frac{1}{2} \text{Tr}_{s,c} [B \Gamma (D_\nu S) \Gamma_{\text{src}}] - \frac{1}{2} \text{Tr}_{s,c} [(B \overleftarrow{D}_\nu) \Gamma S \Gamma_{\text{src}}]. \quad (7.9)$$

The `opt_einsum` expression for the local part,

```
"wtzyxabij,gbn,wtzyxncji,ca->gwtzyx"
```

closes backward color/spin (a, b, i, j) , gamma (b, n) , forward line (n, c, j, i) , and source gamma (c, a) in the order of Eq. (7.8). Fourier projection then supplies the $+\mathbf{q}$ insertion phase. The same contraction with $D_\nu S$ and $B \overleftarrow{D}_\nu$ gives Eq. (7.9).

7.4 Connected proton EMT contraction

For the proton, the raw flavor-specific sequential propagator already contains K_u or K_d . The local primitive can be written

$$L_{\Gamma,f}^p(x; t_f) = \text{Tr}_{s,c} [B_{\text{seq},f}(x; t_f) \Gamma S_f(x; t_f)], \quad (7.10)$$

with the identical right/left derivative construction of Eq. (7.9). In `proton_EMT_vibe_develop.py`, `_raw_seq_gamma_scalar_fields` contracts the complex-conjugated raw sequential line with the active propagator and then with $\gamma_5 \Gamma$. This is the raw-sequential representation of the same $\gamma_5 S_{\text{seq}}^\dagger \gamma_5 \Gamma S$ trace. The up/down Wick multiplicities remain those of Eq. (5.14).

7.5 Stochastic disconnected quark EMT

The local stochastic primitive is

$$T_\Gamma(x) = \xi^\dagger(x) \Gamma \eta(x), \quad \eta = D^{-1} \xi, \quad (7.11)$$

which estimates $+\text{Tr}[\Gamma S]$. For the derivative operator, production computes a right-acting central difference and reconstructs the left-acting piece with gamma-five Hermiticity and momentum reversal. If $\Gamma^\sharp = \gamma_5 \Gamma^\dagger \gamma_5 = s_\Gamma \Gamma_{\text{partner}}$, the stored derivative is

$$L_{\Gamma,\nu}^{\text{disc}}(q) = -\frac{1}{4} [A_R(\Gamma, \nu; q) - s_\Gamma A_R(\Gamma_{\text{partner}}, \nu; -q)^*]. \quad (7.12)$$

The factor in Eq. (7.12) combines the central derivative, the two-sided $\overleftrightarrow{D}$ normalization, and the physical closed-fermion-loop minus. Thus the derived disconnected EMT derivative has the loop sign included. By contrast, the separately stored primitive local bilinear remains the positive trace estimator and must not be interpreted as $\langle \bar{q} \Gamma q \rangle$ without Eq. (3.5).

7.6 Gradient flow and ringed normalization

At positive flow time, gauge and fermion fields obey schematically

$$\partial_t B_\mu = D_\nu G_{\nu\mu}, \quad \partial_t \chi = \Delta \chi, \quad \partial_t \bar{\chi} = \bar{\chi} \overleftarrow{\Delta}, \quad (7.13)$$

with $B_\mu(0) = A_\mu$, $\chi(0) = q$, and $\bar{\chi}(0) = \bar{q}$. Flowed local products are finite building blocks, but flowed fermions require the appropriate ringed-field normalization, and a physical EMT is obtained by a small-flow-time expansion. Schematically,

$$T_{\mu\nu}^R(x) = \lim_{t \rightarrow 0} \left[c_g(t) \tilde{\mathcal{O}}_{\mu\nu}^g(t, x) + \sum_f c_f(t) \tilde{\mathcal{O}}_{\mu\nu}^{qf}(t, x) + \text{trace and mixing operators} \right]. \quad (7.14)$$

The standalone ringed helper estimates the kinetic normalization input

$$K(t) = \langle \bar{\chi}(t, x) \overleftrightarrow{D} \chi(t, x) \rangle \quad (7.15)$$

through the zero-momentum diagonal derivative sum. In the production normalization convention,

$$K_{\text{pervec}}(t, \tau) = -\frac{2}{V_3} \sum_{\mu=1}^4 L_D[\gamma_\mu, \mu, q=0; t, \tau]. \quad (7.16)$$

This is an input to a ringed factor, not the factor itself, and is not automatically multiplied into connected or disconnected EMT output.

7.7 Flowed gluon primitive

The clover field strength is constructed from four oriented plaquettes:

$$F_{\mu\nu}^{\text{clover}} \propto -i \left[Q_{\mu\nu} - Q_{\mu\nu}^\dagger - \frac{1}{N_c} \text{Tr}_c(Q_{\mu\nu} - Q_{\mu\nu}^\dagger) \mathbf{1} \right]. \quad (7.17)$$

The code then measures

$$G_{\mu\nu}^{\text{code}}(t, x) = 2 \text{Tr}_c \sum_{\rho} F_{\mu\rho}(t, x) F_{\nu\rho}(t, x). \quad (7.18)$$

Equation (7.18) is a useful flowed primitive. It lacks the $\delta_{\mu\nu} F_{\rho\sigma} F_{\rho\sigma}$ trace structure, the physical Euclidean sign/coefficient convention, and the matching/mixing coefficients required for a complete gluon EMT.

7.8 EMT audit interpretation

The right/left derivative, symmetrization, and lattice 1/4 normalization are implemented consistently for the connected pion/proton and derived disconnected quark EMT. The output is nevertheless *unringed/unrenormalized flowed data*. Trace combinations, the ringed factor, small-flow matching, quark–gluon mixing, and a controlled window

$$a^2 \ll 8t \ll \Lambda_{\text{QCD}}^{-2} \quad (7.19)$$

belong to the subsequent analysis.

8 qTMD/qGTMD and qPDF/qGPD

8.1 Hadron bilocal matrix element

A convenient translation-covariant definition of the bare coordinate-space matrix element is

$$\widetilde{W}_\Gamma^f(b; P, \Delta, \eta) = \langle H(p_f) | \bar{q}_f(-b/2) \Gamma \mathcal{W}_\eta[-b/2, b/2] q_f(b/2) | H(p_i) \rangle. \quad (8.1)$$

The code uses an equivalent basepoint/end-point form,

$$\mathcal{O}_\Gamma^f(x; b) = \bar{q}_f(x) \Gamma \mathcal{W}[x, x+b] q_f(x+b). \quad (8.2)$$

Translation between Eqs. (8.1) and (8.2) only changes the known phase associated with the operator center; it does not change the Wilson path.

For $b = b_z \hat{z} + b_T \hat{e}_T$ and a staple of extent η , the forward case $\Delta = 0$ is a qTMD coordinate-space matrix element. When $\Delta \neq 0$, it is a qGTMD matrix element. The straight-link limit $b_T = 0$ is the qPDF matrix element for $\Delta = 0$, and the qGPD matrix element for $\Delta \neq 0$.

8.2 Straight Wilson line and covariant transport

For positive b_z ,

$$W[x, x + b_z \hat{z}] = U_z(x) U_z(x + \hat{z}) \cdots U_z(x + (b_z - 1) \hat{z}). \quad (8.3)$$

For negative displacement, the reversed conjugate link product is required:

$$W[x, x - b \hat{z}] = U_z^\dagger(x - \hat{z}) U_z^\dagger(x - 2\hat{z}) \cdots U_z^\dagger(x - b \hat{z}), \quad b > 0. \quad (8.4)$$

Repeated application of Eq. (7.4) produces

$$(\mathcal{T}_{+z}^{b_z} q)(x) = W[x, x + b_z \hat{z}] q(x + b_z \hat{z}) \quad (8.5)$$

for positive b_z , and Eq. (8.4) for negative b_z . The qPDF implementation restarts the negative branch from the undisplaced field at $b_z = -1$, so every incremental step is nearest neighbor.

8.3 Fixed-length GI staple

The GI qTMD index is

$$[b_T, b_z, \eta, \text{dir}_T]. \quad (8.6)$$

For even b_z , nonnegative b_T , and $\eta \geq |b_z|/2$, production decomposes the path into signed segments

$$[(z, \eta + b_z/2), (e_T, b_T), (z, b_z/2 - \eta)]. \quad (8.7)$$

The net endpoint displacement is

$$(\eta + b_z/2)\hat{z} + b_T\hat{e}_T + (b_z/2 - \eta)\hat{z} = b_z\hat{z} + b_T\hat{e}_T, \quad (8.8)$$

and, for the allowed range,

$$L_{\text{path}} = |\eta + b_z/2| + b_T + |b_z/2 - \eta| = 2\eta + b_T. \quad (8.9)$$

The implementation applies the segment list in reverse order. This follows operator composition:

$$\mathcal{T}_{s_1}\mathcal{T}_{s_2}\mathcal{T}_{s_3}q \quad \text{is evaluated by applying } \mathcal{T}_{s_3} \text{ first.} \quad (8.10)$$

Because every $\mathcal{T}$ pulls an endpoint field back to the base point, the reverse execution order is the correct physical staple, not a reversed Wilson line.

At $b_T = 0$ and $\eta = |b_z|/2$, Eq. (8.7) collapses to the positive or negative straight link. Odd b_z is excluded by this half-integer-free fixed-length parameterization, not by a fundamental restriction on Wilson lines.

8.4 Gauge covariance and CG versus GI

Under a local gauge transformation,

$$q(x) \rightarrow \Omega(x)q(x), \quad \bar{q}(x) \rightarrow \bar{q}(x)\Omega^\dagger(x), \quad U_\mu(x) \rightarrow \Omega(x)U_\mu(x)\Omega^\dagger(x + \hat{\mu}). \quad (8.11)$$

Every internal $\Omega^\dagger\Omega$ cancels in a link product:

$$W[x, y] \rightarrow \Omega(x)W[x, y]\Omega^\dagger(y). \quad (8.12)$$

Therefore

$$\bar{q}(x)\Gamma W[x, y]q(y) \rightarrow \bar{q}(x)\Omega^\dagger(x)\Gamma\Omega(x)W[x, y]\Omega^\dagger(y)\Omega(y)q(y), \quad (8.13)$$

which is invariant because color transformations commute with spin Γ . Random-SU(3) tests of the production straight and staple transporters agree with this proof to 3.38×10^{-15} .

The **GI_*** paths use these covariant transports. The **CG_*** paths use ordinary coordinate shifts and omit W . They are meaningful only after a specified gauge fixing, such as Coulomb gauge. A filename containing **cg** is not a mathematical enforcement of gauge fixing; the method, tolerance, and convention must be provenance metadata.

8.5 Connected pion qTMD/qGTMD and qPDF/qGPD

The pion connected nonlocal correlator is the master expression Eq. (4.13). The shifted active propagator is

$$S_u^{(b)}(x, x_0) = W[x, x + b]S_u(x + b, x_0). \quad (8.14)$$

Substitution into Eq. (4.18) yields

$$C_{3,u}^{\pi,\Gamma}(b) = \sum_x \Phi_q(x) \text{Tr}_{s,c} \left[B_{\text{seq}}(x) \Gamma S_u^{(b)}(x, x_0) \Gamma_{\text{src}} \right]. \quad (8.15)$$

contract_qTMD_GI builds a cached transporter for every staple and applies Eq. (8.14); **contract_PDF** incrementally applies the straight shifts. Both then call the common pion gamma scan, so the spin-color Wick closure is identical to the verified pion C_2/C_3 kernel.

8.6 Connected proton qTMD/qGTMD and qPDF/qGPD

For each active flavor, the proton connected expression is Eq. (5.17) with the same shifted line Eq. (8.14). The down and up sequential sources are respectively the exact two-term and four-term cuts of Eq. (5.6). The production contraction

$$(B_{\text{seq},f})_{ji}^{cf} \Gamma_{im}(S_f^{(b)})_{mj}^{fc} \quad (8.16)$$

closes the correct active flavor. The Wilson geometry and the baryon Wick determinant are logically separate: correctness of one does not compensate for an error in the other. Both pass independent checks here.

8.7 Forward versus off-forward production kinematics

From Eqs. (2.9)–(2.11):

$$\mathbf{q} = 0 \Rightarrow p_i = p_f, \Delta = 0 \quad (\text{qTMD or qPDF}), \quad (8.17)$$

$$\mathbf{q} \neq 0 \Rightarrow p_i = p_f - q, \Delta = q \quad (\text{qGTMD or qGPD}). \quad (8.18)$$

At the audited commit, the pion and proton TMD production runners hard-code $\mathbf{p}_f = 0$. Consequently:

$$p_i = -q, \quad P = -q/2. \quad (8.19)$$

For the forward list, $q = 0$ also gives $P_z = 0$. It is a valid raw zero-momentum bilocal correlator but not the large- P_z quasi-distribution regime required for LaMET matching. Momentum-boosted smearing cannot change this Fourier-projected external momentum. Nonzero- q output is a low- or zero-average-momentum off-forward matrix element and should be labelled qGTMD/qGPD with P, Δ, ξ, t recorded.

8.8 From a bare bilocal to a distribution

The measured coordinate-space object requires, schematically,

$$\widetilde{W}_\Gamma^R = Z_\Gamma^{-1}(b, \eta, a, \mu, \text{scheme}) \widetilde{W}_\Gamma^{\text{bare}}, \quad (8.20)$$

followed by the relevant Fourier transform and quasi-to-light-cone matching. For qTMD/qGTMD, a compatible soft subtraction and rapidity prescription are also required:

$$\widetilde{F}^{\text{sub}} \sim \frac{\widetilde{W}^R}{\sqrt{S^R(b_T, \eta)}}, \quad \frac{\partial \ln \widetilde{F}}{\partial \ln \sqrt{\zeta}} = K(b_T, \mu). \quad (8.21)$$

The soft function, rapidity scale ζ , Collins–Soper kernel K , and LaMET matching kernel are not harmless overall constants. The current HDF5 correlators are bare inputs to such an analysis, not physical TMDs, GTMDs, PDFs, or GPDs.

9 Disconnected qTMD/PDF loop and vacuum subtraction

9.1 Operator trace produced by the code

For the same straight or staple operator, production solves $\eta = D^{-1}\xi$, transports η to $\mathcal{W}\eta$, and contracts

$$T_\Gamma(b, q) = \sum_x \Phi_q(x) \xi^\dagger(x) \Gamma \mathcal{W}[x, x+b] \eta(x+b). \quad (9.1)$$

Its noise expectation is

$$\mathbb{E}_\xi[T_\Gamma] = + \text{Tr} \left[\Phi_q \Gamma \mathcal{W}_b D^{-1} \right]. \quad (9.2)$$

This agrees with the stored metadata:

```
loop_convention = xi_dagger_Gamma_0_b_eta
trace_target    = Tr[P_qtau Gamma 0_b Dinv]
```

9.2 Physical Wick loop

The physical bilinear loop follows from Eq. (3.5):

$$L_{\text{phys}}(b, q) = - \text{Tr} \left[\Phi_q \Gamma \mathcal{W}_b D^{-1} \right] = -\mathbb{E}_\xi[T_\Gamma]. \quad (9.3)$$

For a hadron two-point measurement C_2^H , the disconnected three-point contribution is the vacuum-subtracted covariance

$$\begin{aligned} C_{3,\text{disc}}^{\text{phys}} &= \langle C_2^H L_{\text{phys}} \rangle - \langle C_2^H \rangle \langle L_{\text{phys}} \rangle \\ &= - \left[\langle C_2^H T_\Gamma \rangle - \langle C_2^H \rangle \langle T_\Gamma \rangle \right]. \end{aligned} \quad (9.4)$$

Vacuum subtraction does not remove or alter the global minus.

9.3 Confirmed sign issue and data consequence

The qTMD/PDF loop producer correctly stores the positive trace estimator in Eq. (9.2). The accompanying physical-combination description at the audited commit uses the positive covariance without the outer minus in Eq. (9.4). No Wilson-path or gamma convention supplies a compensating sign. This is a **confirmed P0 physical-sign issue**.

For current-schema raw data with explicit convention metadata, the correction is exact:

$$\text{physical_loop} = -\text{loop_pervec}. \quad (9.5)$$

Inversions need not be repeated; any already constructed positive-trace covariance must be multiplied by -1 and recomputed with the desired vacuum subtraction. Legacy files without sufficient provenance cannot be automatically classified by their gamma channel alone.

The disconnected EMT derivative is not affected by this particular omission: its derived quantity explicitly includes the minus in Eq. (7.12). Its primitive local trace still follows the positive-trace warning.

10 Pion qTMD wave function and quasi-distribution amplitude

10.1 Vacuum-to-pion bilocal

The sink bilocal realized by the qTMDWF/qDA contraction is

$$\mathcal{O}_\Gamma(x; b) = \bar{d}(x+b) W[x+b, x] \Gamma u(x). \quad (10.1)$$

With a local pion source $\bar{u}(x_0) \Gamma_{\text{src}} d(x_0)$, flavor-preserving Wick pairing gives

$$C_\Gamma(x; b) \propto \text{Tr}_{s,c} [S_d(x_0, x+b) W[x+b, x] \Gamma S_u(x, x_0) \Gamma_{\text{src}}], \quad (10.2)$$

with the same pseudoscalar source-adjoint convention as Eq. (4.8). Its spectral limit is Eq. (2.14).

10.2 Why shift before dagger gives the correct endpoint

Production first shifts or transports the source-to-sink backward-flavor propagator:

$$\tilde{S}_d(x, x_0) = W[x, x + b] S_d(x + b, x_0). \quad (10.3)$$

It then forms the backward line:

$$\begin{aligned} \gamma_5 \tilde{S}_d(x, x_0)^\dagger \gamma_5 &= \gamma_5 S_d(x + b, x_0)^\dagger W[x, x + b]^\dagger \gamma_5 \\ &= S_d(x_0, x + b) W[x + b, x]. \end{aligned} \quad (10.4)$$

The Wilson line is color-only and commutes with γ_5 . Substitution of Eq. (10.4) into the common pion trace produces Eq. (10.2). Thus the shift–dagger order does not reverse the physical operator; it creates the required antiquark endpoint and conjugate link.

10.3 qDA as the straight bT=0 limit

The gauge-invariant qDA restricts $b_T = 0$ and uses the straight transporter of Eqs. (8.3)–(8.4). A common leading-twist continuum definition is

$$\begin{aligned} \langle 0 | \bar{d}(z/2) \gamma_z \gamma_5 W[z/2, -z/2] u(-z/2) | \pi(P) \rangle &= i f_\pi P_z \int_0^1 dx e^{i(x-\frac{1}{2})z P_z} \phi_\pi(x, \mu) \\ &+ \text{higher twist}, \end{aligned} \quad (10.5)$$

up to Euclidean continuation and normalization conventions. Therefore the axial channels, such as Z5 or T5 for a specified quasi-DA choice, carry the leading-twist DA interpretation. A scan of all 16 gamma matrices is useful operator data but is not sixteen independent leading-twist DAs.

The GI qDA endpoint and straight path are correct. The CG qDA and the current qTMDWF implementation use ordinary shifts and require fixed-gauge input. Both remain bare vacuum-to-pion matrix elements until nonlocal renormalization and matching are performed. A gauge-invariant qTMDWF would also require a staple and soft/rapidity scheme compatible with the target factorization theorem.

11 Legacy pion soft-factor four-point measurement

11.1 Wall-source propagators

For a Coulomb-gauge wall source at t_0 and quark momentum $\mathbf{k}$,

$$\eta_{\mathbf{k}}(\mathbf{x}, t) = \delta_{t, t_0} e^{+i\mathbf{k} \cdot \mathbf{x}}, \quad G_{\mathbf{k}}(x; t_0) = D^{-1} \eta_{\mathbf{k}}. \quad (11.1)$$

The legacy contraction uses both $+\mathbf{k}$ and $-\mathbf{k}$ wall propagators and constructs backward antiquark lines with gamma-five Hermiticity.

11.2 Four-propagator Wick blocks

For source time t_0 and sink separation t_{sep} , define

$$G_w = G_{k_{\text{fw}}}(t_0), \quad G_{w, b^\dagger} = G_{-k_{\text{bw}}}(t_0), \quad (11.2)$$

$$G_{w, \dagger} = G_{-k_{\text{fw}}}(t_0 + t_{\text{sep}}), \quad G_{w, b} = G_{k_{\text{bw}}}(t_0 + t_{\text{sep}}). \quad (11.3)$$

After the legacy momentum-transfer phase on the sink-side backward propagator and an ordinary transverse shift by b , the two closed spin-color blocks are

$$A_b(x) = G_w(x) \Gamma_{\text{src}} \gamma_5 G_{w,b^\dagger}(x+b)^\dagger \gamma_5, \quad (11.4)$$

$$B_b(x) = G_{w,b}(x+b) \Gamma_{\text{sink}} \gamma_5 G_{w,\dagger}(x)^\dagger \gamma_5. \quad (11.5)$$

The measured four-point correlator is

$$C_4(t; t_{\text{sep}}, b, \Gamma_1, \Gamma_2) = \sum_x \text{Tr}_{s,c} [A_b(x) \Gamma_2 B_b(x) \Gamma_1]. \quad (11.6)$$

The production `tmp_1`, `tmp_2`, and final `einsum` implement Eqs. (11.4)–(11.6) in this order. The legacy pion four-propagator Wick algebra is therefore internally consistent.

11.3 What the four-point contraction does not prove

Equation (11.6) is a pion wall-source four-point function with ordinary fixed-gauge transverse shifts. A universal TMD soft function is normally a vacuum expectation value of Wilson-line geometry defined in the same rapidity regulator and renormalization scheme as the unsubtracted TMD. Correct contraction of Eq. (11.6) alone does not prove

$$C_4 \equiv S_{\text{vacuum}}^{\text{scheme}} \quad (11.7)$$

or prove that dividing the production GI staple matrix element by a ratio built from C_4 performs the desired soft subtraction. That identification requires an explicit factorization derivation matching staple length, directions, rapidity regulator, gauge convention, and renormalization scheme.

12 Background-current and current–current response

12.1 First derivative of a propagator

Let

$$D(\lambda) = D + \lambda O, \quad S(\lambda) = D(\lambda)^{-1}. \quad (12.1)$$

Differentiating $D(\lambda)S(\lambda) = 1$ gives

$$\left. \frac{dS}{d\lambda} \right|_0 = -SOS. \quad (12.2)$$

For a local current with momentum transfer,

$$O_q(x) = \Gamma_{\text{current}} \Phi_q(x; x_0), \quad (12.3)$$

the inserted source is

$$\eta_O(x) = O_q(x) S(x, x_0), \quad (12.4)$$

and its inversion yields

$$D^{-1} \eta_O = SOS. \quad (12.5)$$

`invert_local_current_response_propagator` constructs the positive quantity in Eq. (12.5) by default. It is appropriate for a direct comparison to an explicitly summed insertion. Interpreting it as the derivative of $D + \lambda O$ requires the minus in Eq. (12.2); the helper exposes this as `response_sign`.

If only a source-relative time window $\mathcal{T}$ is selected, the source is

$$\eta_O^\mathcal{T} = \sum_{\tau \in \mathcal{T}} \delta_{x_4, t_0 + \tau} O_q(x) S(x, x_0). \quad (12.6)$$

The one inversion then produces the insertion summed over that window. It is not an unsummed $C_3(\tau)$.

12.2 Ordered second response

For $D(\lambda_1, \lambda_2) = D + \lambda_1 O_1 + \lambda_2 O_2$,

$$\left. \frac{\partial^2 S}{\partial \lambda_2 \partial \lambda_1} \right|_0 = SO_2 SO_1 S + SO_1 SO_2 S. \quad (12.7)$$

The nested helper constructs the single ordered term

$$SO_2 SO_1 S. \quad (12.8)$$

It is a current–current response diagnostic or one time ordering, not by itself the full mixed derivative in Eq. (12.7). A physical second-order analysis must define the second ordering, time windows, coincident-current contact terms, and the spectral decomposition independently of a direct three-point formula.

13 Audit evidence and formula-to-code verdicts

13.1 Independent CPU oracles

The numerical oracles use explicit NumPy sums and do not import the production contraction functions. Their double-precision algebraic tolerance was 10^{-11} .

Oracle	Formula tested	Maximum discrepancy/result
Pion Wick oracle	Full gamma-five backward line and $C_2^\pi = \text{Tr}(S^\dagger S)$	5.68×10^{-14}
Proton Wick oracle	Complete flavor-labelled direct–exchange determinant and $T_2^{\text{code}} = -\mathcal{X}$	5.55×10^{-14} for full C_2
Sequential oracle	$K_d = \partial C_2 / \partial S_d$ and $K_u = \partial C_2 / \partial S_u$	1.99×10^{-15} , 2.08×10^{-14}
Production adapter	Actual production down/up kernels and pion backward line versus independent reference	3.55×10^{-15} , 3.56×10^{-15} ; pion exact
Disconnected oracle	Complete-basis trace estimator, determinant loop sign, and covariance	Trace error 4.48×10^{-16} ; physical covariance exactly opposite
Geometry oracle	Positive/negative straight link, staple chain, plane-wave derivative, Fourier phase, and SU(3) covariance	3.38×10^{-15}

13.2 Single-A100 S8T8 evidence

On a single A100 with `mpi_geometry=1.1.1.1`, the audit collected 63 relevant tests: 61 passed and two were skipped only because optional pre-generated comparison directories were absent. A real HYP-smeared *S8T8* gauge field confirmed the cached GI staple against stepwise `covDev`, including positive and negative b_z , transverse staples, and the PDF limit, at 10^{-12} tolerance.

The source and pion-sequential solves reached true residuals 4.77×10^{-13} and 7.47×10^{-13} , respectively. Pion and proton C_2 , pion EMFF at $q = 0, 1$, GI qGTMD, GI qGPD, CG qTMDWF, GI qDA, and the symmetric derivative all produced finite arrays.

A four-rank run was not required to resolve any remaining formula conflict: the compound-index, phase, and Wilson-path questions were already decided by explicit oracles and real-gauge single-rank tests. This is not a claim of single-/multi-rank bitwise equivalence; a future MPI boundary regression should compare 1.1.1.1, 1.1.2.2, and 2.2.1.1 when that parallel property is the target.

13.3 Source-level map

Production location	Formula represented	Verdict
pion_utils_vibe_develop.py:154--157	$S(x_0, x) = \gamma_5 S(x, x_0)^\dagger \gamma_5$ with full endpoint reversal	Correct
pion_utils_vibe_develop.py:236--246	$BT_{\text{sink}} ST_{\text{src}}$ spin-color trace and Fourier sum	Correct
proton_utils_vibe_develop.py:87--176	Proton direct-exchange determinant	Correct ; printed term 2 is $-\mathcal{X}$
bw_seq_pyquda.py:143--176	Meson fixed-sink source $\Phi_{pf} \gamma_5 \Gamma_{\text{sink}}^\dagger \gamma_5 S$	Correct
bw_seq_pyquda.py:181--300	Down/up proton sequential cuts	Correct ; two/four terms
pion_EMFF_vibe_develop.py:172--190	One-active-line local-current pion C_3	Correct raw correlator , not F_π
qtmd_operator_utils.py:102--222	Straight and fixed-length staple transport	Correct geometry
application/nucleon_TMD/shared_runner.py:230--260	Proton nonlocal insertion trace	Correct raw connected trace
pion_qTMDWF_pyquda.py:130--240	Shifted backward line for qDA/qTMDWF	Correct endpoint ; CG needs gauge fixing
flowed_fermion_bilinear_vibe_develop.py:148--181	Central derivative and left-line reconstruction	Correct lattice derivative
Disconnected_1pt_EMT_vibe_develop.py:302--329	Two-sided derivative loop	Closed-loop sign included
Disconnected_1pt_qTMD_vibe_develop.py:85--160	Positive stochastic qTMD/PDF trace	Physical loop needs an outer minus
Disconnected_1pt_EMT_vibe_develop.py:1025--1110	Clover F and $2 \text{Tr } FF$	Correct primitive , not complete EMT
pion_soft_factor_vibe_develop.py:208--280	Legacy two-block pion C_4	Correct specified C_4 ; universal soft identification unproved
pion_current_background_response_vibe_develop.py:49--175	SOS and ordered $SO_2 SO_1 S$	Correct diagnostics ; derivative sign must follow definition

14 Measurement-by-measurement conclusions and data impact

Measurement	Verdict	Physics qualification	Existing-data consequence
Pion common C_2/C_3	Correct	Flavor path, gamma-five backward line, and spin-color trace agree.	No Wick-based recomputation required.
Proton common C_2	Correct	Full u -line direct-exchange determinant; no u/d cross contraction.	No recomputation from the exchange concern.
Proton up/down sequential	Correct	Exact flavor derivatives of the same C_2 ; up multiplicity already included.	Do not multiply the up channel by two again.
Pion EMFF	Conditionally correct	Raw one-line local-current C_3 ; missing complete physical extraction.	Retain C_3 ; add missing C_2 , charges, Z_V , improvement, and ratio.
Proton EMFF	Not a dedicated target	Common local insertion exists, but no full F_1/F_2 workflow.	Do not relabel generic local-limit output as F_1, F_2 .
Connected quark EMT	Conditionally correct	Derivative and Wick primitives correct; flowed and unrenormalized.	Retain primitives; perform trace basis, ringed normalization, matching, mixing.
Disconnected quark EMT	Conditionally correct	Derived derivative includes the loop sign; primitive local trace does not.	Use dataset-level convention metadata.
Gluon EMT	Conditionally correct	$2 \text{ Tr } FF$ primitive only.	Retain primitive; construct trace/matching basis in analysis.
Connected GI qTMD/qPDF	Correct raw kernel	Wick and path correct; current forward runner has $P_z = 0$, with no soft/matching.	Raw data valid, but not a LaMET distribution; large- P_z target requires new data.
Connected qGTMD/qGPD	Correct raw kernel	$q \neq 0$ is off-forward with $P = p_f - q/2$, $\Delta = q$.	Retain raw data and label/store P, Δ, ξ, t .
Disconnected qTMD/PDF	P0 sign issue	Saved trace is positive; physical Wick loop is negative.	Current-schema raw loops can be exactly sign-corrected without new inversions.
qTMDWF	Conditionally correct	Endpoint Wick path correct; current implementation is CG/fixed-gauge.	Usable only with gauge-fixing provenance and later renormalization/-matching.
qDA	Conditionally correct	GI $b_T = 0$ straight endpoint correct; leading twist needs axial channel.	Bare GI correlator can be retained for later analysis.

Measurement	Verdict	Physics qualification	Existing-data consequence
Pion soft C_4	Specified algebra correct	No demonstrated equality to the GI quasi-TMD vacuum soft function.	Retain as a legacy C_4 ; do not divide into GI qTMD without factorization proof.
Background response	Correct diagnostic	Positive SOS versus derivative $-SOS$ is explicit; nested helper is one ordering.	Interpret using response-sign and time-window metadata.

15 Compact derivation checklist for new analyses

For every output channel, the following checks should be completed in order:

1. Write the continuum operator and hadron matrix-element decomposition. State flavor charges and external-state normalization.
2. Decide whether the file contains a raw correlator, fitted bare matrix element, or renormalized/matched observable.
3. Use the actual plus-sign **MomentumPhase** convention to derive $p_i = p_f - q$. Store P, Δ, ξ, t for off-forward data.
4. Preserve flavor labels in the Wick derivation. Apply $S_u = S_d$ only after forbidden cross-flavor contractions have been excluded.
5. For a proton, verify the complete direct-exchange determinant and use a flavor derivative to define every sequential source.
6. For a disconnected bilinear, distinguish the positive stochastic trace from the negative physical closed loop before vacuum subtraction.
7. For a bilocal, prove the endpoint and Wilson-link orientation after every dagger; distinguish fixed-gauge CG from gauge-invariant GI.
8. For flowed EMT data, keep trace subtraction, ringed normalization, small-flow matching, and quark-gluon mixing explicit.
9. Only after these steps construct ratios, fits, renormalization, matching, soft subtraction, and continuum/flow-time limits.

16 Final physics statement

Starting from physical interpolators and preserving field order and flavor, the connected pion and proton production kernels implement the intended Wick paths. In particular:

$$C_2^\pi = \text{Tr}_{s,c}(S^\dagger S) \quad (16.1)$$

in the equal-line pseudoscalar limit,

$$C_2^p = -\mathcal{D} + \mathcal{X} \quad (16.2)$$

contains the full same-flavor u determinant, and

$$K_f = \frac{\partial C_2^p}{\partial S_f} \quad (16.3)$$

generates the correct proton sequential sources. Straight and staple transporters have the intended endpoints, and shifting the qTMDWF/qDA backward-flavor line before gamma-five Hermitian conjugation yields the correct conjugate Wilson line.

The one confirmed contraction-level physical problem is

$$\langle \bar{q} \Gamma \mathcal{W} q \rangle = -\text{Tr}(\Gamma \mathcal{W} D^{-1}), \quad (16.4)$$

whereas the disconnected qTMD/PDF producer saves the positive stochastic trace and its documented physical covariance omits the outer minus. This is an exact convention error, not a statistical or solver effect.

All other major qualifications in this note concern the distinction between a correct raw lattice building block and a final observable: EMFF requires charge/current normalization and a ratio or spectral fit; EMT requires the full flowed matching and mixing program; and nonlocal parton observables require suitable external momentum, renormalization, matching, and, for TMDs, a compatible soft and rapidity treatment.

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